

Exercise13

CPS I — Exercise Sheet 13

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Exercise 1: LTL Properties

(a)satisfies

- $\{a\{b^\omega$
- $\{a\{a\{b^\omega$
- $\{c\{b^\omega$
- $\{c\{a^\omega$
- $\{a^\omega$
- $\{c^\omega$

(b)violates

- $\{a^\omega$
- $\{a\{c^\omega$
- $\{a\{b^\omega$
- $\{a^\omega$
- $\{b^\omega$
- $\{c\{a^\omega$

(c)TS

- No, because $A_0 = \{b$ or $A_0 = \{a, c$ $A_1 = \{a$ dont satisfy a and next b
- Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$
- No, because $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$ satisfies $aU \Box b$
- Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$
- Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$
- Yes, infinite c. $(s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow)^\omega$

(d)set comprehension

- $Words(\phi_1) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid a \in A_0 \wedge b \in A_1\}$
- $Words(\phi_2) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \exists k \in N. (b \in A_k \wedge \forall i < k. a \in A_i)\}$
- $Words(\phi_3) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \neg(\exists k \in N. (\forall j \geq k. b \in A_j) \wedge (\forall i < k. a \in A_i))\}$

- $Words(\phi_4) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \exists k \in N. (c \in A_k \wedge \forall i > k. a \in A_i)\}$
- $Words(\phi_5) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \exists k \in N. \forall i \geq k. a \in A_i\}$
- $Words(\phi_6) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \forall k \in N. \exists i \geq k. a \in A_i\}$

Exercise 2: Stating properties in LTL

- (a) $\Box \neg (\text{Peter.use} \wedge \text{Betsy.use})$
 (b) $\Box (\text{Peter.use} \rightarrow \Diamond \text{Peter.release}) \wedge \Box (\text{Betsy.use} \rightarrow \Diamond \text{Betsy.release})$
 (c) $\Box (\text{Peter.request} \rightarrow \Diamond \text{Peter.use}) \wedge \Box (\text{Betsy.request} \rightarrow \Diamond \text{Betsy.use})$
 (d) $\Box (\text{Peter.request} \rightarrow \Diamond \neg \text{Peter.request}) \wedge \Box (\text{Betsy.request} \rightarrow \Diamond \neg \text{Betsy.request})$
 (e) $\Box ((\text{Peter.use} \rightarrow (\neg \text{Peter.use} \cup \text{Betsy.use})) \wedge (\text{Betsy.use} \rightarrow (\neg \text{Betsy.use} \cup \text{Peter.use})))$

Exercise 3: From Set Notation to LTL

- (a) $\phi_1 = \Diamond (a \wedge \bigcirc b)$
 (b) No
 (c) $\phi_3 = \Box ((a \leftrightarrow \bigcirc \bigcirc a) \wedge (b \leftrightarrow \bigcirc \bigcirc b))$
 (d) $\phi_4 = \Box (a \rightarrow \Diamond b)$